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Received: 11 January 2024 | Accepted: 15 July 2024

DOI: 10.1002/ajb2.16388

SPECIAL ISSUE ARTICLE

Botany

The role of deep hybridization in fern speciation: Examples from the Thelypteridaceae

Yu-Hsin Tseng[1] | Li-Yaung Kuo[2] | Israel Borokini[3,4] | Susan Fawcett[4,5]

[1]Department of Life Sciences, National Chung Hsing University, no. 145 Xingda Rd., South District, Taichung 40227, Taiwan

[2]College of Life Science, National Tsing Hua University, No. 101, Section 2, Kuang Fu Road, Hsinchu 30044, Taiwan

[3]Department of Ecology, Montana State University, 310 Lewis Hall, Bozeman, MT 59717, USA

[4]University and Jepson Herbaria, University of California, Berkeley, 1001 Valley Life Sciences Building, Berkeley, CA 94720-2465, USA

[5]National Tropical Botanical Garden, 3530 Papālina Road, Kalāheo, HI 96741, USA

Correspondence

Susan Fawcett, University and Jepson Herbaria, University of California, Berkeley, 1001 Valley Life Sciences Building, Berkeley CA 94720-2465 USA.
Email: susan.fawcett@gmail.com

Abstract

Premise: Hybridization is recognized as an important mechanism in fern speciation, with many allopolyploids known among congeners, as well as evidence of ancient genome duplications. Several contemporary instances of deep (intergeneric) hybridization have been noted, invariably resulting in sterile progeny. We chose the christelloid lineage of the family Thelypteridaceae, recognized for its high frequency of both intra- and intergeneric hybrids, to investigate recent hybrid speciation between deeply diverged lineages. We also seek to understand the ecological and evolutionary outcomes of resulting lineages across the landscape.

Methods: By phasing captured reads within a phylogenomic data set of GoFlag 408 nuclear loci using HybPhaser, we investigated candidate hybrids to identify parental lineages. We estimated divergence ages by inferring a dated phylogeny using fossil calibrations with treePL. We investigated ecological niche conservatism between one confirmed intergeneric allotetraploid and its diploid progenitors using the centroid, overlap, unfilling, and expansion (COUE) framework.

Results: We provide evidence for at least six instances of intergeneric hybrid speciation within the christelloid clade and estimate up to 45 million years of divergence between progenitors. The niche quantification analysis showed moderate niche overlap between an allopolyploid species and its progenitors, with significant divergence from the niche of one progenitor and conservatism to the other.

Conclusions: The examples provided here highlight the overlooked role that allopolyploidization following intergeneric hybridization may play in fern diversification and range and niche expansions. Applying this approach to other fern taxa may reveal a similar pattern of deep hybridization resulting in highly successful novel lineages.

KEYWORDS

allopolyploidy, *Christella*, hybrid speciation, intergeneric hybrid, niche overlap, polyploidy, reticulate evolution

The importance of hybridization in speciation, especially in disturbed or novel habitats, has long been recognized (Stebbins, 1940, 1950; Anderson and Stebbins, 1954). Among ferns, thought to have fewer reproductive barriers to hybridization than angiosperms (Smith, 1972; Ranker and Sundue, 2015), over 1200 instances of inter- and intrageneric hybrids have been reported (Knobloch, 1996). Polyploidization is a critical mechanism that enables the restoration of fertility and offers an escape from the evolutionary dead end typical of most hybrids. This form of speciation is prevalent among ferns; a change in ploidy is associated with 31% of speciation events (Wood et al., 2009), half of which are estimated to have resulted from allopolyploidization (Barker et al., 2016). A steady accrual of morphological, cytological, and molecular

This article is part of joint special issues of the *American Journal of Botany* and *Applications in Plant Sciences*: "Twice as Nice: New Techniques and Discoveries in Polyploid Biology."

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evidence has deepened our appreciation for the importance of allopolyploid speciation among ferns, which has been well documented among recently diverged congeners (Manton, 1950; Barrington, 2020), whereas few examples of fertile homoploid hybrids in ferns have been documented (e.g., Walker, 1958; Hickok and Klekowski, 1973; Conant and Cooper-Driver, 1980; Chao et al., 2022). In addition, ancient genome duplication events followed by diversification are inferred to have occurred more than 18 times among ferns (Leebens-Mack et al., 2019; Pelosi et al., 2022), suggesting that polyploidy may play a fundamental role in shaping diversity.

The ecological underpinning of hybridization is long established in literature. Anderson (1949) hypothesized that the survival of hybrids is associated with availability of disturbance-created habitat conditions intermediate between those of the progenitors, which prevents parents from outcompeting biologically fragile progenies (Folk et al., 2013). Previous studies have shown that potential for hybridization between closely related taxa increases when both taxa have higher similarity in their ecological niches (Andújar et al., 2014; but see Rueda et al., 2021) and have sympatric distributions (Liao et al., 2015; Zhang et al., 2018); however, frequency of hybridization increases under marginal ecological conditions (Song et al., 2011; Ortego et al., 2014). The ecological niche of an allopolyploid may be expanded, contracted, or novel, relative to its diploid progenitors; however, the most frequently documented pattern is niche overlap and geographic sympatry, suggesting that fine-scale ecological factors may be important for niche differentiation (Glennon et al., 2014; Marchant et al., 2016; Baniaga et al., 2020). Fixed heterozygosity has been suggested as a mechanism for the success of allopolyploids by stabilizing the hybrid phenotype, buffering them against inbreeding and drift (Stebbins, 1985; Brochman et al., 2004), and increasing their ecological amplitude, or tolerance of drought and temperature extremes (Hijmans et al., 2007). Stressful conditions and the availability of novel habitats are associated with the success and persistence of allopolyploids relative to progenitor species, especially among ferns (Barrington, 2020; Van de Peer et al., 2021). Consequently, anthropogenic activities have caused disturbances that enhance hybridization in many plant taxa (Grabenstein and Taylor, 2018), and provide additional opportunities for the formation and success of allopolyploids.

In general, capacity for hybridization decreases with increasing genetic distance between parents, although this relationship may be nonlinear, and highly variable among lineages (Stebbins, 1950, Moyle et al., 2004; Abbott et al., 2013; Sigel, 2016). Hybridization is a common phenomenon among closely related fern congeners, but rare examples of hybridization between lineages that diverged from each other many millions of years ago have been reported (reviewed by Liu et al., 2020a). However, not all of these putative hybrids are supported by the same standard of evidence. Among those investigated using cytological and/or molecular data, ×*Osmunimunda ruggii* (Wagner et al., 1978), a hybrid between *Osmunda spectabilis* and *Claytosmunda claytoniana*—estimated to have diverged 98 million years ago (Ma)—is the example of deepest known parental divergence. Other notable examples of hybrids between parents of similarly deep divergence are reported in the Osmundaceae (Tsutsumi et al., 2012), Dryopteridaceae (Wagner et al., 1992), and Lindsaeaceae (Lehtonen, 2018) based on a recent fossil-calibrated dated phylogeny (Nitta et al., 2022). We use the term deep hybridization to refer to a cross between highly divergent lineages, usually represented by distinct genera, which serve as an imperfect but useful taxonomic proxy for divergence (Schuettpelz et al., 2018). These aforementioned cases of deep (intergeneric) hybridization are recent events resulting in highly sterile first-generation progeny and do not represent examples of ancient hybridization sensu Stull et al. (2023), which refers to past formation of stable evolutionary lineages.

Typically, hybridization is inferred through indicators such as intermediate morphologies, overlapping ranges of potential parent species, increased chromosome number in allopolyploids, and cytonuclear discordance. However, these indicators fade over thousands to millions of years of evolution (Stull et al., 2023). Recent significant progress in phylogenomics via next-generation sequencing (NGS) data has enhanced our ability to detect both recent and ancient hybridization events (e.g., Folk et al., 2017; Rothfels, 2021; Mendez-Reneau et al., 2023). Among the phylogenomic methods, targeted capture (Hyb-Seq) has emerged as a powerful technique, which utilizes DNA or RNA baits to enrich pre-identified loci to generate extensive sequence data for phylogenomics (Andermann et al., 2020). Universal probe kits for target capture, such as Angiosperms353 for angiosperms (Johnson et al., 2019) and Goflag 451 and Goflag 408 for flagellate land plants (Breinholt et al., 2021), have revolutionized plant evolution studies, enabling sequencing of hundreds of nuclear genes across broad phylogenetic groups. Novel bioinformatic pipelines such as Hybphaser (Nauheimer et al., 2021) and PATÉ (Tiley et al., 2021) can phase alleles from polyploids using target-capture sequence data to detect hybridization or introgression events, enriching our understanding of the role of hybridization in plant evolution.

STUDY SYSTEM

Perhaps no other group of ferns is so prone to hybridization as is the genus *Christella*. The taxonomic treatment of *Christella* of China (Li et al., 2013) accepted 22 species and judged an additional 22 named taxa to be putative hybrids based on morphological intermediacy, irregular fronds, and malformed spores. *Christella* has also proven to be a cooperative subject for experimental study of hybridization, and many of its species readily form hybrids in a greenhouse setting (Panigrahi and Manton, 1958; Ghatak et al., 1971). Based on the cytological evidence resulting from this work, the widespread and variable diploid species *Christella hispidula* is a likely genomic contributor to some of the most widespread and weedy tetraploid species, including *Christella dentata*, which itself has been implicated in many intrageneric and intergeneric hybrids (Table 1). Hybridization experiments clarified that the taxon

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TABLE 1 List of previously reported F_1 intergeneric hybrids in the christelloid clade.

Progenitor 1	Progenitor 2	Hybrid	Reference
Amblovenatum opulentum	*Abacopteris aspera*	×*Chrismatopteris holttumii*	Quansah and Edwards (1986)
Amblovenatum opulentum	*Strophocaulon invisum*	NA	Game and Smith (2014)
Christella dentata	*Amblovenatum opulentum*	*Amblovenatum tildeniae*	Holttum (1977); Game and Smith (2014)
Christella dentata	*Grypothrix triphylla*	×*Chrisnephrium insulare*	TPG (2019)
Christella dentata	*Pneumatopteris afra*	*Thelypteris dentata* var. *buchananii*	Holttum (1974)
Christella dentata	*Menisciopsis cyatheoides*	×*Christelliopsis palmeri*	Wagner (1993)
Christella dentata	*Menisciopsis penangiana*	*Thelypteris* ×*nepalensis*	Fraser-Jenkins (2008b)
Christella hispidula	*Pneumatopteris afra*	*Thelypteris* ×*varievenulosa*	Viane (1985)
Christella parasitica	*Grypothrix triphylla*	*Thelypteris* ×*thwaitesii*	Nayar and Geeverghese (1987)
Christella sp.	*Abacopteris nudata*	*Thelypteris* ×*jerdonii*	Holttum (1976)
Christella procera	*Grypothrix triphylla*	*Thelypteris* ×*gogoii*	Fraser-Jenkins (2008a)
Christella queenslandica	*Abacopteris aspera*	×*Abacopterella altifrons*	Almeida et al. (2023)
Glaphyropteridopsis erubescens	*Menisciopsis penangiana*	*Thelypteris* ×*occulta*	Fraser-Jenkins et al. (2017)
Glaphyropteridopsis erubescens	*Pseudocyclosorus esquirolii*	×*Glaphyrocyclosorus erubesquirolicus*	Shieh and Tsai (1987)
Pneumatopteris afra	pseudocyclosoroid	*Thelypteris* ×*wildemanii*	Viane (1985)

now recognized as *C. hispidula* is represented by individuals from Borneo, Ghana, Nigeria, and Sri Lanka that were fully interfertile. Further evidence for the coherence of this variable pantropical species was provided by a phylogenomic analysis (Fawcett et al., 2021) that included accessions from Brazil, Bolivia, the Dominican Republic, Ethiopia, Indonesia, Mexico, and the southeast United States, which together form a clade.

In addition to an abundance of intrageneric hybrids, especially within the genus *Christella*, there are also a surprising number of sterile intergeneric hybrids, most treated taxonomically as nothogenera, reported for this group (Table 1). Putative intergeneric hybrids include 15 combinations involving members of the christelloid clade, 10 of which include *Christella* as a progenitor. The christelloid clade is defined as a predominantly paleotropical group of 19 genera (Fawcett and Smith, 2021; Fawcett et al., 2021).

In his monograph of neotropical members of the christelloid clade, Smith (1971a) postulated a hybrid origin for the tetraploid species that we now recognize as *Pelazoneuron kunthii* and suggested that its diploid progenitors are *Pelazoneuron ovatum* and *Christella hispidula*. Evidence for its hybrid origin included intermediate morphology and meiotic chromosomal evidence from an F_1 hybrid between *P. kunthii* and *P. ovatum* yielding 36 bivalents and 36 univalents, consistent with *P. ovatum* representing a constituent genome of *P. kunthii*.

Because of the astonishing number of natural hybrids reported from the christelloid clade, and the striking patterns of gene tree discordance seen within this group (Fawcett and Smith, 2021; Fawcett et al., 2022), we chose to focus on this group to investigate the potential for deep hybrid speciation. We selected the *Pelazoneuron kunthii* complex for further investigation into niche differentiation because of detailed prior research into its origin and parentage (Smith, 1971a, b) and the abundance of reliably determined specimens available in herbaria that could be used in spatial analyses.

In the present study, we sought to investigate the potential contribution of deep (intergeneric) hybridization to fern diversity by (1) using genomic evidence to infer hybrid origin by phasing homeologous alleles, (2) estimating divergence times among progenitor lineages using a fossil calibrated phylogeny, and (3) investigating climatic niche dynamics among a widespread allopolyploid and its progenitors. We combined this evidence to understand potential drivers and evolutionary outcomes of deep hybridization (Diniz-Filho et al., 2010; Peixoto et al., 2017).

MATERIALS AND METHODS

Sampling design

Our data set of 114 accessions was drawn from a larger published nuclear DNA data set of the Thelypteridaceae using the targeted enrichment GoFlag 408 probe set (Breinholt et al., 2021; Fawcett et al., 2021, 2022; Appendix S1). The present data set includes 114 accessions, with representatives from major lineages within the Thelypteridaceae selected as the outgroup. Clade references were chosen based on sequence quality and to represent putative progenitors. Our sample targets members of our focal lineage, the christelloid clade, with the inclusion of all suspected and known hybrids, and emphasizing the genera *Abacopteris* (6 accessions), *Amblovenatum* (3),

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Christella (31), *Menisciopsis* (8), *Pelazoneuron* (16), and *Pseudocyclosorus* (11). Initial exploration for potential hybrids was done by visually assessing topologies of individual unphased gene trees and assessing concordance factors (Minh et al., 2020). Additional outgroup taxa, *Lorinseria areolata* C. Presl and *Diplazium hymenodes* (Mett.) Á. Löve & D. Löve, were selected from other fern families in Aspleniinae published by Breinholt et al. (2021).

Quality control, trimming, and assembly

The quality of the raw reads was assessed using FastQC v0.12.0 (Andrews, 2010). To ensure high data quality, we trimmed the reads using Trimmomatic v0.39 (Bolger et al., 2014) with a sliding window of 20 bp, accepting only those with quality scores of Q20 or better and a minimum read length of 36 bp. Subsequently, we employed the HybPiper v.2.1.2 pipeline (Johnson et al., 2016) in conjunction with BWA v0.7.17 (Li and Durbin, 2010) to map and align the target regions of the GoFlag 451 bait set (Breinholt et al., 2021). In this study, we opted for the GoFlag 451 bait set instead of GoFlag 408 as the reference to maximize the potential loci obtained. The mapped reads were further assembled into contigs using Exonerate v.2.4.0 (Slater and Birney, 2005). In addition to retrieving the on-target sequences (exon regions), we extracted the off-target sequences (intron regions) with the HybPiper script intronerate.py. Finally, we combined the retrieved exon and intron regions into concatenated sequences for each locus using the setting supercontig of the script retrieve_sequences.py.

Hybrid detection

The pipeline HybPhaser (Nauheimer et al., 2021) was used to detect putative paralogs, assess allele divergence (AD, proportion of SNPs across all loci), locus heterozygosity (LH, proportion of loci that contain SNPs), and phase haplotypes for potential hybrid detection. Default settings of HybPhaser were used, which required a minimum coverage of 10× for single nucleotide polymorphism (SNP) calling, an allele frequency of at least 0.15, and a minimum occurrence of the alternative allele in four reads. Additionally, loci with SNP frequencies greater than 1.5 times the interquartile range compared to other loci were identified as paralogous and excluded from the data set. Loci represented by fewer than three samples were also removed to reduce missing data.

As suggested by Nauheimer et al. (2021), we selected samples that were evenly distributed across the phylogeny and exhibited low AD, low LH, and high coverage of target sequences as clade references (Table 2; Appendix S2). These clade references were *Abacopteris birii* (Clade E), *Amblovenatum opulentum* (accession from Vietnam; Clade F), *Christella papilio* (Clade A), *Christella hispidula* var. *versicolor* (Clade B), *Menisciopsis cyatheoides* (*Wood 16205*; Clade G), *Pelazoneuron ovatum* var. *lindheimeri* (Clade H), *Pneumatopteris humbertii* (Clade C), and *Pseudocyclosorus tylodes* (Clade D; Appendix S3). A total of eight samples were selected for haplotype phasing: samples with high AD and LH that mapped to more than one clade reference (Table 2; Appendix S2) and samples identified as potential hybrids in previous studies (*Christella conspersa*, *Christella chaseana*, *Christella gueintziana*, ×*Christelliopsis palmeri*, *Pelazoneuron patens* var. *dissimile*, *Pelazoneuron kunthii*, *Sphaerostephanos elatus*; Table 3).

Phylogenetic inference and dating

A subset of 107 samples and 359 loci (higher than 99%, ≥106 samples contained sequences) were selected for our phylogenomic matrix representing clades of interest and omitting taxa with long branches and higher proportions of missing data. To align the coding (i.e., exons) and noncoding (i.e., introns) regions of these loci individually, we used MACSE v2.03 (Ranwez et al., 2011) with the setting -max_refine_iter 3 -local_realign_init 0.3 -local_realign_dec 0.2) and MAFFT v7.450 (Katoh and Standley, 2013) with the setting --maxiterate 16 --globalpair, respectively. The program trimAl v1.4.rev22 (Capella-Gutiérrez et al., 2009) was then used with the default setting to trim the noncoding alignments. Resulting alignments including coding and noncoding regions were concatenated into a supercontig matrix. To perform phylogenomic analyses, we first coded this matrix with four partitions (script codon_property) representing the three codon positions and the noncoding regions and assigned a GTR + F + R10 model to each of them. Our maximum likelihood (ML) phylogeny was inferred using IQ-TREE v2.1.3 (Nguyen et al., 2015; Minh et al., 2020), with 1000 ultrafast bootstrap (UFBS) replicates (Hoang et al., 2018). Based on this ML tree, we then inferred the gene and site concordance factors (gCF and sCF; Minh et al., 2020) for each branch.

To infer a dated phylogeny, we used treePL (Smith and O'Meara, 2012) and followed the approach by Maurin (2020). In this study, we used the exon-only ML phylograms and age calibrations on two nodes: the crown node of Thelypteridaceae and the stem node of the cyclosoroid clade (Figure 1). To reduce effects of missing data on phylogenetic branch inference for our treePL dating, we generated new ML phylograms based on the exon data set, after removing 20 exons that contained a high proportion of missing data. Ten tips with unusually long branches in our ML phylogeny (e.g., *Pakau pennigera* in Figure 1) were also removed, because they appeared to contain considerable erroneous bases in the HybPiper supercontigs and/or a high proportion of missing data, which could result in overestimated branch lengths and ages. The included exons and taxa are summarized in Appendix S4. We used IQ-TREE v2.1.3 to first generate ML phylograms using the codon_property partition and model selection (see above). The crown age of Thelypteridaceae was based on previously dated trees (Testo and Sundue, 2016; Nitta et al., 2022), and its maximum and minimum ages were set to 104.43 and

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TABLE 2 Results from the HybPhaser analysis. Putative hybrids are indicated in boldface, and results typical of other accessions are also included for comparison. Note the high level of heterozygosity and allele divergence among putative deep hybrids (indicated in boldface). Scores showing conflicting affinities with different reference clades typical of hybrids are also marked in boldface. Complete results of the analysis are available in Appendix S7.

			Reads matching to each clade reference (%)								
Sample	**Allele divergence (%)**	**Locus heterozygosity (%)**	***Amblovenatum opulentum***	***Christella papilio***	**Christella hispidula var. versicolor**	***Pseudocyclosorus tylodes***	***Abacopteris birii***	***Pneumatopteris humbertii***	***Menisciopsis cyatheoides* (Fawcett)**	***Pelazoneuron ovatum* var. *lindheimeri***	**Note**
Christella chaseana	**2.1**	**99.5**	1.4	3.6	**26.5**	**13.3**	1.9	2.8	1.2	1.1	Phased sample
Christella conspersa	**3.5**	**99.5**	1.3	7.4	**26.7**	1.1	1.5	0.8	1.7	**13.9**	Phased sample
Christella gueintziana	**2.2**	**99.8**	1.4	3.1	**25.3**	**12.3**	1.6	2.5	1.1	1.1	Phased sample
×Christelliopsis palmeri	**3.0**	**100.0**	1.6	7.3	**17.0**	1.5	1.5	0.9	**19.6**	1.3	Phased sample
Pelazoneuron kunthii	**3.5**	**99.8**	0.9	2.4	**26.8**	0.8	0.8	0.6	1.9	**27.2**	Phased sample
Pelazoneuron patens* var. *dissimile	**2.8**	**99.5**	0.9	1.5	**17.7**	0.7	0.8	0.7	1.9	**19.6**	Phased sample
Pelazoneuron schizotis	**3.5**	**99.5**	1.0	2.4	**26.3**	0.8	0.9	0.6	2.1	**27.3**	Phased sample
Sphaerostephanos elatus	**2.0**	**98.5**	1.9	2.2	1.3	**14.6**	**11.9**	3.9	1.7	1.3	Phased sample
Abacopteris birii	0.5	74.4	0.9	2.2	0.8	2.1	46.8	1.0	1.8	0.9	Clade reference
Amblovenatum opulentum Vietnam	0.1	13.2	74.0	0.4	0.2	0.4	0.7	0.4	0.9	0.7	Clade reference
Christella hispidula var. *versicolor*	0.2	35.4	0.9	2.7	45.2	0.8	0.8	0.6	0.7	1.1	Clade reference
Christella papilio	0.1	14.2	0.7	55.6	1.9	0.9	0.6	0.5	1.2	1.1	Clade reference
Menisciopsis cyatheoides Fawcett	0.2	45.6	1.1	0.5	0.4	0.5	0.6	0.5	67.6	1.0	Clade reference
Pelazoneuron ovatum var. *lindheimeri*	0.2	41.5	0.4	0.3	0.2	0.4	0.3	0.1	0.7	80.6	Clade reference
Pneumatopteris humbertii	0.4	82.3	0.9	0.8	0.4	1.7	0.6	59.6	0.7	0.6	Clade reference
Pseudocyclosorus tylodes	0.1	26.1	0.5	0.7	0.3	60.0	0.5	2.4	1.0	0.8	Clade reference

(Continues)

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TABLE 2 (Continued)

Sample	Allele divergence (%)	Locus heterozygosity (%)	Reads matching to each clade reference (%)								Note
			Amblovenatum opulentum	*Christella papilio*	*Christella hispidula* var. *versicolor*	*Pseudocyclosorus tylodes*	*Abacopteris birii*	*Pneumatopteris humbertii*	*Menisciopsis cyatheoides* (Fawcett)	*Pelazoneuron ovatum* var. *lindheimeri*	
Christella hispidula Ethiopia	3.8	99.2	1.2	2.6	25.1	1.1	1.2	1.0	2.7	5.2	
Grypothrix rubicunda	0.9	87.9	2.5	2.8	1.6	2.8	2.7	2.0	16.7	5.5	
Menisciopsis boydiae	0.2	15.7	1.6	0.9	0.4	0.8	0.8	0.7	61.6	1.2	
Menisciopsis lakhimpurensis	0.4	63.5	2.4	2.4	1.5	2.0	1.8	1.3	29.6	2.9	
Strophocaulon unitum	0.4	72.2	5.7	5.2	3.9	4.5	5.7	2.6	5.6	2.0	

86.63 Myr. For the stem node of the cyclosoroid clade, we set the minimum age to 66 Myr based on the earliest cyclosoroid fossils (Naugolnykh et al., 2016; Berry, 2019; Xu et al., 2019). We also applied the same treePL setting to the 1000 UFBS phylograms to infer the age ranges while accounting for uncertainty in branch length and topology. To verify that the models did not greatly influence the inferred ages, we carried out the same dating analyses on the phylograms inferred using different partitions and substitution models (Appendix S4), which allow for other phylogenetic uncertainties. To infer the best partition scheme, we used ModelFinder (Kalyaanamoorthy et al., 2017) and the Bayesian information criterion (BIC) with the setting -rcluster 10.

Verifying maternal parentage of intergeneric hybrids

For hybrids without published plastid DNA sequences, we mapped their GoFlag reads with the plastome of *Christella parasitica* (L.) H. Lév. ex Y.H. Chang (GenBank accession MT130643; Du et al., 2021) using Geneious (Kearse et al., 2012). Their consensus sequences at the plastid regions *rbcL*, *trnL-F*, and *rps4* were used to BLASTn against the nucleotide collection in GenBank. Given that plastids in ferns are presumably inherited from the maternal lineage (reviewed by Kuo et al., 2018), we could confirm the maternal origins of these intergeneric hybrids by identifying the parental genera with the highest sequence similarity.

Quantification of ecological niche differences

We investigated whether the ecological niches of the two progenitors and the hybrid species, *Pelazoneuron kunthii*, are more similar than expected by chance. We hypothesize that *Pelazoneuron ovatum* and *Christella hispidula* have significantly similar ecological niches, which would result in the opportunity for hybridization of the two diploid ferns in areas of overlap. To investigate climatic niche differences and test the niche conservatism hypothesis, we adopted the centroid, overlap, unfilling, and expansion (COUE) framework, based on principal component analysis (PCA; Broennimann et al., 2012; Petitpierre et al., 2012). Occurrence data were drawn from the Pteridophyte Collections Consortium (pteridoportal.org) and filtered to georeferenced specimens determined by experts. Additional specimens at UC were georeferenced from label data (Figure 2; Appendix S5). For this pairwise niche difference quantification, the expert-verified occurrence records were used to define the climatic space and their ecologically accessible areas for each species pair in the first two PCA axes of the 19 bioclimatic variables (30 arc seconds resolution) from WorldClim version 2.1 (Fick and Hijmans, 2017), and potential evapotranspiration and global aridity index (Zomer et al., 2022). Accessible areas were defined as the

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TABLE 3 Deep (intergeneric) hybrids among members of the christelloid clade evaluated in this study. Bold type denotes the inferred maternal progenitor based on BLAST results of chloroplast sequences newly generated for this study, except for *Pelazoneuron kunthii*, *Christella conspersa*, and *Sphaerostephanos elatus*, which are based on the results of Patel et al. (2019). The most recent common ancestor estimate is from Nitta et al. (2022) (ftol_ult_con_dated_lab_v.1.1.0), measured in millions of years before present, and from present analysis.

Progenitor 1	Progenitor 2	Hybrid	MRCA (Nitta et al., 2022)	MRCA (this study)
Christella hispidula [2*x*]	Pseudocyclosoroids [2*x*, 3*x*, 4*x*]	*Christella chaseana*	21.1	23.6
Christella hispidula [2*x*]	Pseudocyclosoroids [2*x*, 3*x*, 4*x*]	*Christella gueintziana*	21.1	23.6
Christella hispidula [2*x*]	***Pelazoneuron ovatum*** [2*x*]	*Pelazoneuron kunthii* [4*x*]	31.6	44.5
Christella hispidula [2*x*]	***Pelazoneuron***	*Pelazoneuron schizotis*	31.6	44.5
Christella hispidula inconstans	***Pelazoneuron***	*Pelazoneuron patens dissimile*	31.6	44.5
Christella hispidula [2*x*]	*Pelazoneuron*	*Christella conspersa* [4*x*]	31.6	44.5
Christella dentata [4*x*]	***Menisciopsis cyatheoides*** [2*x*]	×*Christelliopsis palmeri* [3*x*]	27.86	33.6
Abacopteris	Pseudocyclosoroids	*Sphaerostephanos elatus*	21.1	22.2

set of ecoregions with at least five occurrence records of the species (Barve et al., 2011). We used the global terrestrial ecoregions mapped by Dinerstein et al. (2017) to delineate accessible areas (Figure 2A–C). Following this, the gridded climatic spaces were binned into 100 × 100 km cells corresponding to available suitable ecological conditions. A kernel density smoothing function was applied to correct sampling bias and spatial autocorrelation in the species occurrence densities gridded on the climatic space (Broennimann et al., 2012; Petitpierre et al., 2012). We then performed a PCA ordination analysis on the climatic space occupied by each species pair, which was used to assess niche overlap, similarity, and equivalency. We used Schoener's *D* metric to estimate niche overlap, which varies from 0 (no overlap) to 1 (complete overlap). To quantify niche stability, expansion, and unfilling, we extracted niche density values of the gridded climatic space for the pair of species. Niche stability is the proportion of the climatic space occupied by both species used for the analysis. Niche expansion (= 1 – stability) is the portion of the environmental conditions occupied by the second species, but not occupied in the first species, while niche unfilling estimates the portion of the climatic space that is occupied by the first species but not occupied by the second species (Petitpierre et al., 2012; Guisan et al., 2014).

To assess the statistical significance of these niche difference tests, we conducted two randomization tests based on niche similarity and niche equivalency assumptions. The niche similarity test investigates whether randomly sampled ecological niches of the two progenitors are more similar than expected by chance (Pili et al., 2020; Liu et al., 2020b). Taxa that are phylogenetically closely related are expected to have greater niche similarity than expected by chance and hence exhibit niche conservatism (Peterson et al., 1999; Wiens and Graham, 2005; Warren et al., 2008). The niche equivalency test investigates the transferability of niches across time and space and thus assesses whether the ecological niches of two taxa randomly drawn from the same underlying distribution are more different than expected by chance (Warren et al., 2008). If two taxa have undergone niche divergence, their niches will be less equivalent than random expectations (Peterson, 2011). This pattern would be expected when paleoecological conditions drive divergent selection and allow for adaptation to novel environments (Raxworthy et al., 2007; Schluter, 2009). The COUE framework was originally developed to test for niche variation among natural and invasive populations of a species (Broennimann et al., 2012; Petitpierre et al., 2012). Thus, the niche similarity test is more appropriate to test for niche conservatism for recent events such as distributional shifts and for intraspecific taxonomic units (that is, populations) due to shared ancestry (Peterson et al., 1999; Wiens and Graham, 2005; Warren et al., 2008). In this study, however, we are interested in how evolutionary events such as deep hybrid speciation are associated with shared niche properties among the three fern taxa; hence, the niche equivalency test is relevant for drawing inference (Peterson, 2011). Therefore, our null hypothesis is that niches are identical (equivalent) if the niche equivalency test (with low *D* assumption) is nonsignificant, regardless of a significant (or nonsignificant) niche similarity test. In contrast, the niche equivalency hypothesis is not supported, and niche divergence is inferred when the niche equivalency test is significantly different than random *D* values, regardless of the niche similarity test results (Callen and Miller, 2015; Brown and Carnaval, 2019; Vaissi and Rezaei, 2022). For both niche similarity and equivalency tests, we generated 1000 randomizations of null distributions of simulated niche overlap values to compare with the actual *D* (Borokini et al., 2023). We ran these randomizations under both the assumptions of higher and lower niche overlap than expected by chance between a pair of species. This analysis was conducted the R package ecospat version 3.5.1 (Di Cola et al., 2017) in R version 4.2.2 (R Core Team, 2022).

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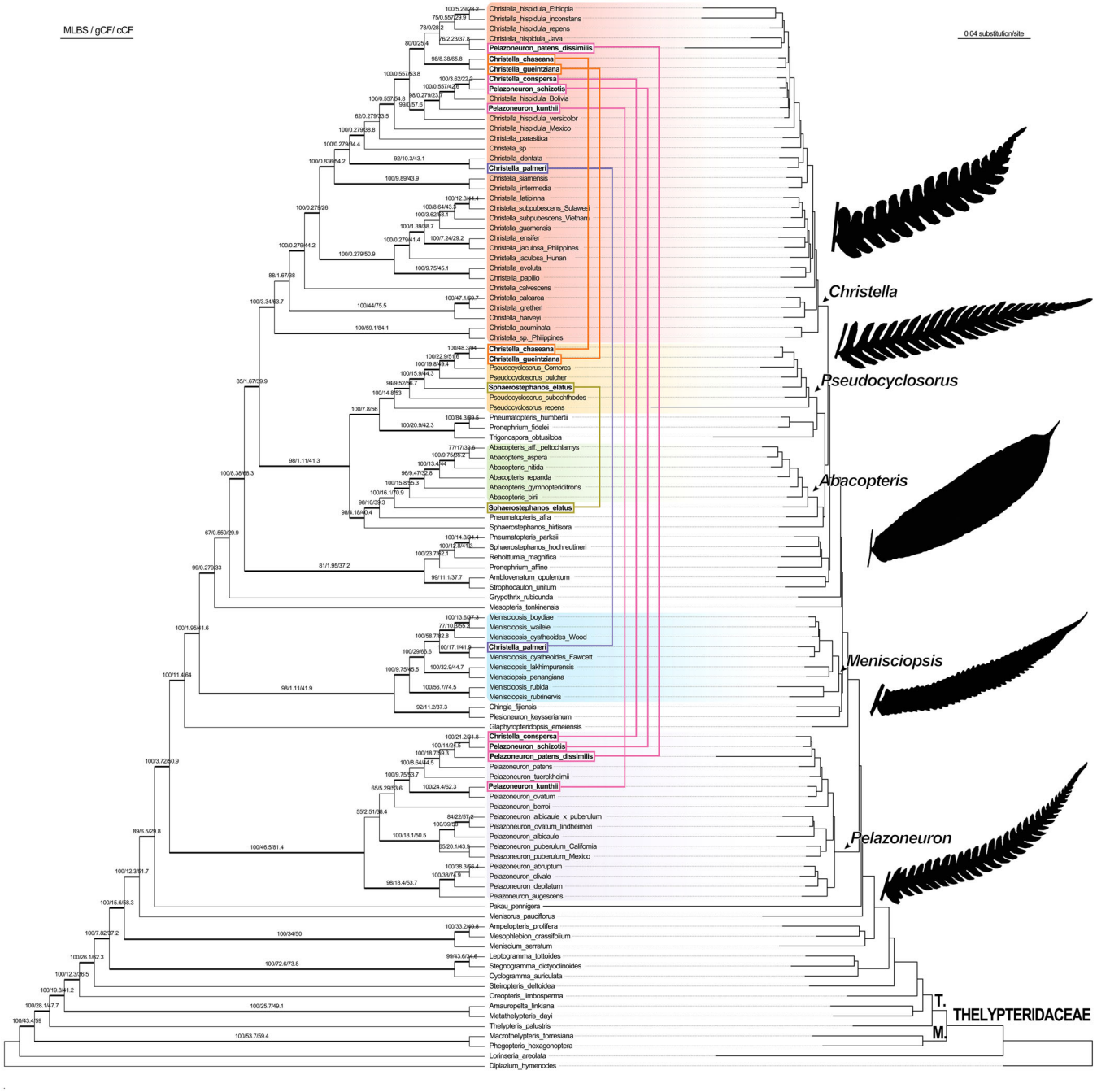


FIGURE 1 Maximum likelihood (ML) phylogeny of the christelloid clade inferred from *GoFlag 408* loci. Phased homeolog assemblies of hybrids are indicated by colored boxes with lines indicating the positions of their homeologous subgenomes. Numbers above branches indicate maximum likelihood bootstrap values (MLBS)/gene concordance factors (gCF)/site concordance factors (sCF).

RESULTS

Hyb-Seq data efficiency

We successfully captured an average of 408 genes per sample from the GoFlag 451 loci, with a range of 215 to 437 genes across the samples. The mean percentage of on-target reads in the christelloid group was 40.4% (22.7–57.2%, Appendix S6). For gene recovery, the exon sequences had an average length of 75,286 bp (range: 44,922–82,368 bp), intron of 257,193 bp (77,294–443,777 bp), and supercontig of 336,096 bp (143,983–518,430 bp). We removed a total 57 putative paralogs and 28 loci represented by fewer than three samples from the initial GoFlag 451 loci data set.

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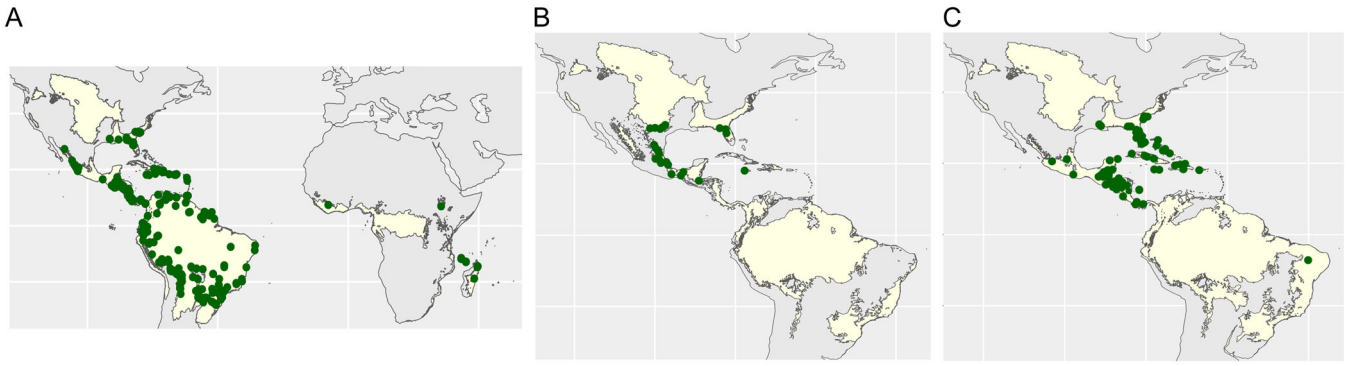


FIGURE 2 Global distribution of (A) *Christella hispidula*, (B) *Pelazoneuron ovatum*, and (C) *P. kunthii* used for the niche overlap analysis. Green dots represent the georeferenced specimens (Appendix S5); yellow-shaded areas indicate the ecologically accessible areas used for the niche overlap analysis.

Further cleaning by default settings in trimAl led to the final data set, which comprised 114 samples and 366 loci used for subsequent analyses.

Assessment of locus heterozygosity, allele divergence, and allele phasing

In general, samples exhibited moderate locus heterozygosity (LH, mean = 69.4%) and allele divergence (AD, mean = 1%) on average, while 12 samples displayed high LH (>98.5%) and high AD (>2%), as detailed in Table 2 and Appendix S2. Among these samples, *Christella hispidula* from Ethiopia had the highest rates of LH (99.2%) and AD (3.8%), and the lowest was in *Amblovenatum opulentum* from Vietnam (13.2% LH and 0.06% AD).

The reads of eight phased samples mapped unambiguously and in high proportion to two reference clades (Figure 1, Table 2; Appendix S7). Five phased samples (*Sphaerostephanos elatus*, *Pelazoneuron patens* var. *dissimile*, ×*Christelliopsis palmeri*, *Pelazoneuron kunthii*, *Pelazoneuron schizotis*) had approximately equal proportions of unambiguous mapped reads to two clades (1:1) and three samples (*Christella chaseana*, *C. gueintziana*, *C. conspersa*) had a proportion of roughly 2:1 (Table 2).

Phylogenomic inference and divergence time estimation

The inferred relationships among genera were largely congruent with the results of earlier studies (He and Zhang, 2012; Almeida et al., 2016; Patel et al., 2019; Fawcett et al., 2021; Figure 3, Table 3). The divergence time estimates were similar to those inferred in a recent study (Nitta et al., 2022; Table 3; Appendix S4), especially for the divergence of *Christella* from the pseudocyclosoroids and of *Abacopteris* from the pseudocyclosoroids; however, the divergence time inferred between *Christella* and *Pelazoneuron* was earlier in our estimate (45 Ma) than in Nitta et al. (2022) (35 Ma).

Ecological niche difference analyses

The PCA-based niche difference quantification revealed moderate ($D = 0.29–0.45$) pairwise overlap in the environmental niches of the three fern taxa, with the allopolyploid hybrid showing greater overlap with each of its progenitors than the overlap between progenitors (Figure 4, Table 4). The randomization tests yielded mixed results for the pairwise niche similarity and equivalency tests (Figure 5, Table 4). Randomizations on the niche equivalency test (with low D assumption) showed significant support for niche divergence between *C. hispidula* and *P. ovatum*, and *C. hispidula and P. kunthii* (Table 4, Figure 4D, E); while the *P. kunthii* and *P. ovatum* niches were no more equivalent than expected by chance (Table 4). In contrast, randomization tests on niche similarity showed a significant difference only between *P. kunthii* and *P. ovatum*, indicating shared niche properties and conservatism (Table 4, Figure 4F).

In all cases, niche overlap among the ferns could be decomposed into high niche stability (0.77–0.99), that is, niche properties shared by both taxa (Table 4, Figure 4A–C). This pattern of niche overlap was also corroborated in the niche centroid plot, which showed a small change (red arrows) within the density of overlapping occupied ranges of both taxa (that is, niches shared by both taxa (Figure 4A–C). Further, niche expansion (niche found in species 2 but not in species 1) was low in all cases, except between *P. kunthii* and *P. ovatum*, where 23% of the niche occupied by the *P. ovatum* was not found in *P. kunthii* (Table 4). Likewise, 21% of the niche occupied by *C. hispidula* was not occupied by *P. ovatum* (unfilling; Table 4, Figure 5).

DISCUSSION

Evidence for intergeneric hybrid speciation

The HybPhaser analysis provided strong evidence supporting a long-standing hypothesis of deep allopolyploid origin

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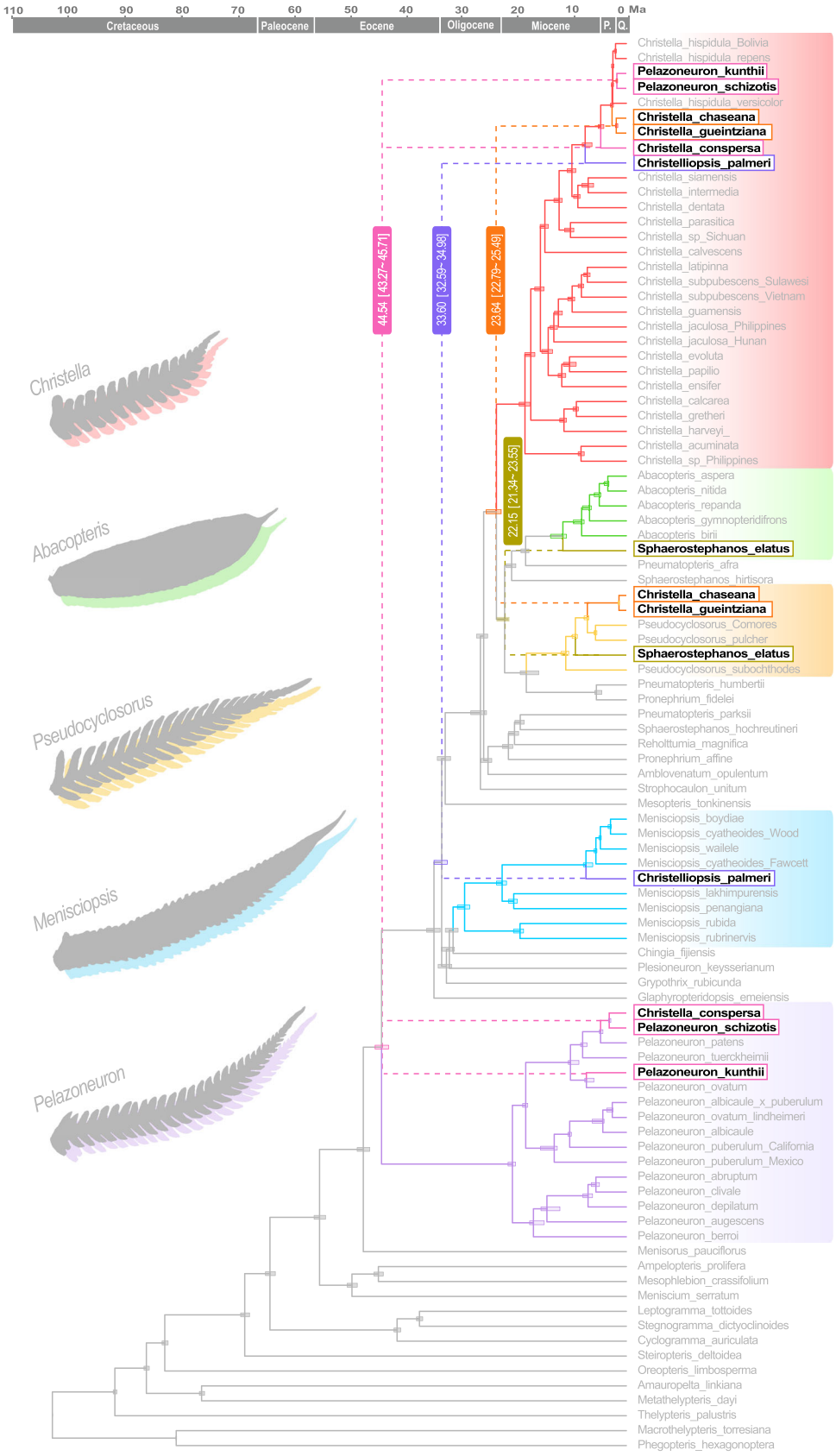


FIGURE 3 Fossil-calibrated chronogram with divergence age estimates. The bars on nodes indicate their 95% highest posterior density (HPD) age intervals from the maximum likelihood bootstrap trees. Colors correspond to generic clades of interest; hybrids are in boldface and outlined with a colored box. Dotted lines correspond to putative hybrids with divergence age estimates between progenitor lineages indicated.

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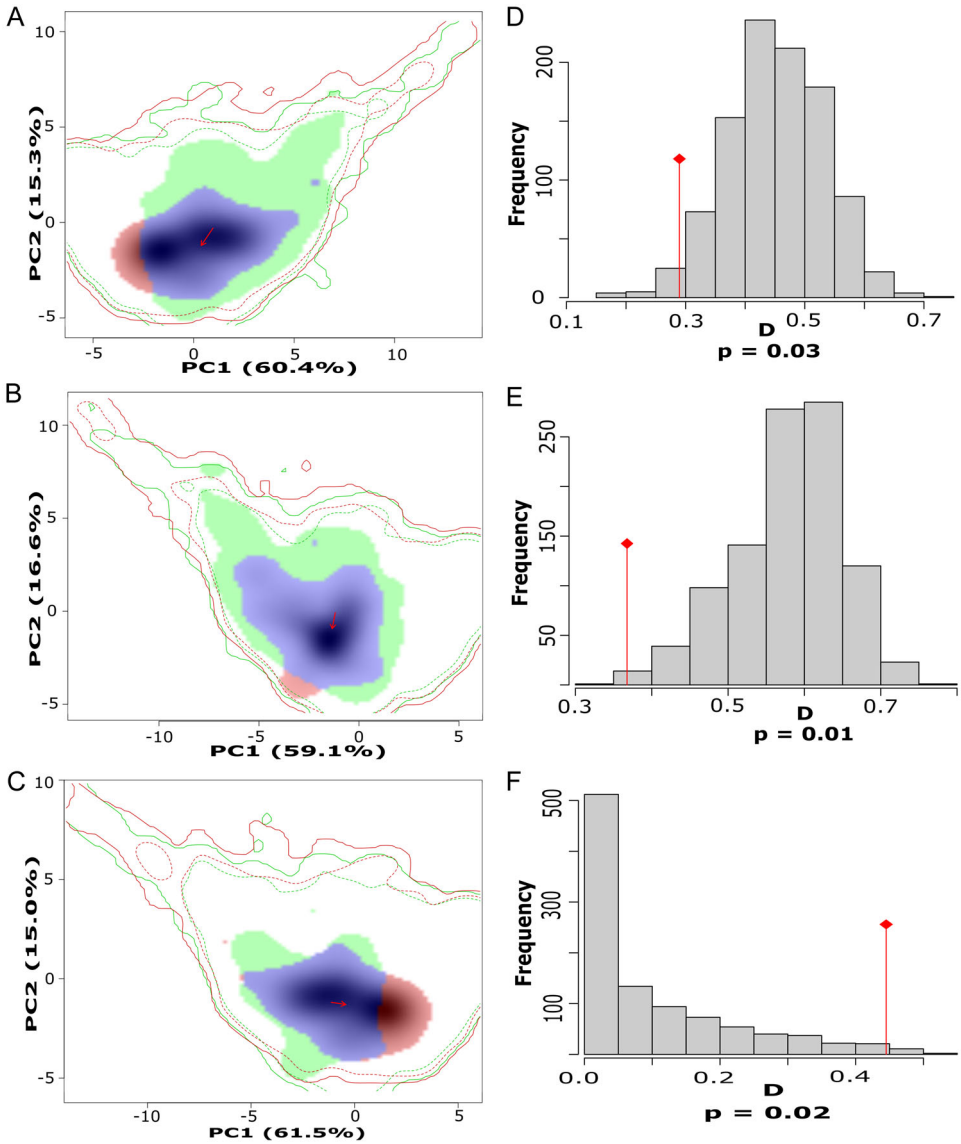


FIGURE 4 Pairwise niche dynamics based on the first two axes of the PCA calibrated on the global distribution ranges of (A) *Pelazoneuron kunthii* and *Christella hispidula*, (B) *P. ovatum* and *C. hispidula*, and (C) *P. ovatum* and *P. kunthii*. The blue-shaded area represents niche overlap between two fern taxa; pink shading represents niches occupied by species 1 but not species 2; green shading indicates ecological conditions persistent for species 2 but not species 1. Solid and dashed contour lines represent the results using, respectively, 100% and 75% of the full range of available background environmental space. The red solid arrow illustrates niche differences from the global environmental niche centroid of species 1 to the niche centroid of species 2; the red dashed arrow indicates the difference at 75% of the niches of a pair of taxa.

TABLE 4 Pairwise niche comparisons among the *Pelazoneuron kunthii* and its progenitors. Niche overlap metric (Schoener's *D*) ranges from 0 (no niche overlap) to 1 (completely similar niches). Tests of significance under niche similarity (high *D* assumption) and equivalency (low *D* assumption) were conducted with 1000 randomizations each. Niche dynamics show the decomposition of niche overlap into stability, expansion, and unfilling estimates. Metrics with $P < 0.05$ are shown in bold font.

		Randomization tests		Niche dynamics		
Taxa pairs*	**Niche overlap *D***	**Niche similarity**	**Niche equivalency**	**Stability**	**Expansion**	**Unfilling**
Christella hispidula × *Pelazoneuron ovatum*	0.286	0.189	**0.027**	0.919[b]	0.081[b]	0.213
C. hispidula × *P. kunthii*	0.400	0.153	**0.006**	0.988	0.012	0.138[b]
P. kunthii × *P. ovatum*	0.445	**0.014**	0.284	0.768[b]	0.232[b]	0.108[a]

*In all cases, the first listed taxon is the reference species in the analysis.

[a]Significant results ($P < 0.05$) in niche similarity test only.

[b]Significant results ($P < 0.05$) in niche equivalency test only.

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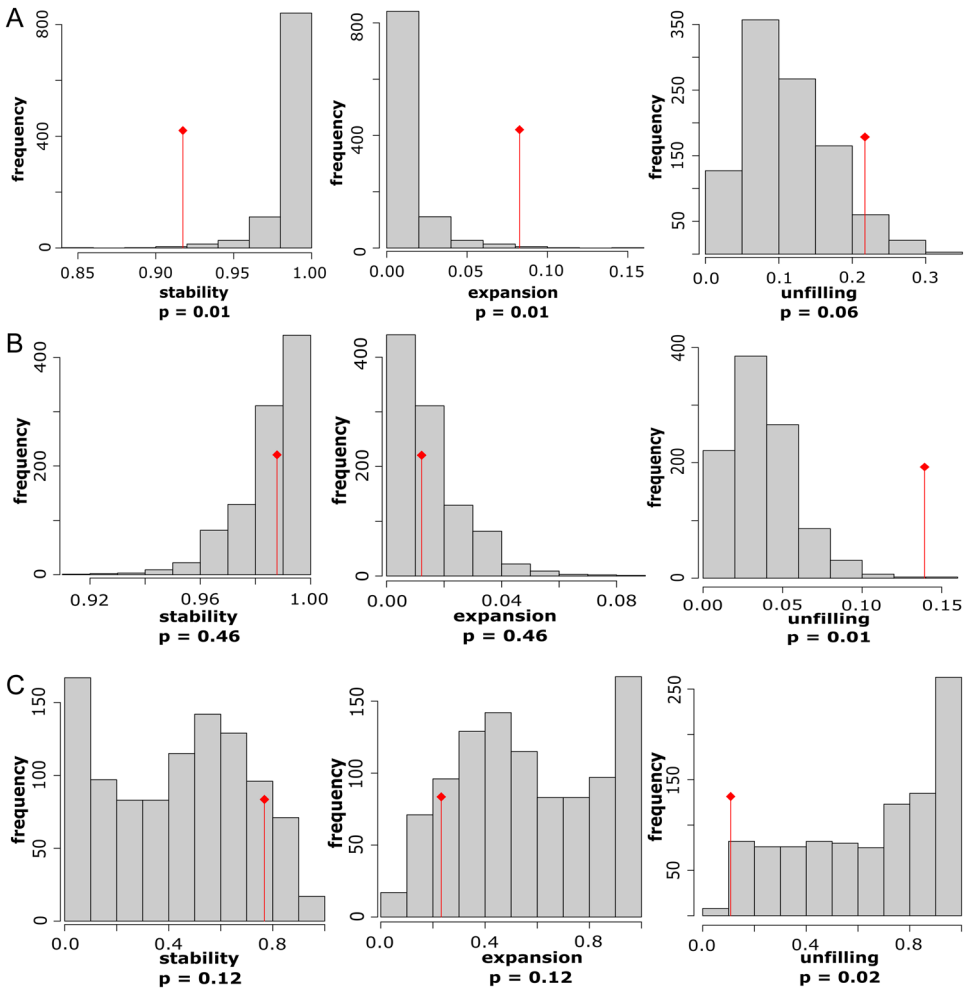


FIGURE 5 Frequency histograms showing the random distributions of Schoener's *D* from the 1000 randomizations compare with the observed *D* statistic (red diamond) for niche similarity tests between (A) *Pelazoneuron kunthii* and *Christella hispidula*, (B) *P. ovatum* and *C. hispidula*, and (C) *P. ovatum* and *P. kunthii*.

of *Pelazoneuron kunthii*, a common, widespread species, and revealed previously unreported examples of deep hybridization within the christelloid group (Figure 1; Tables 1–3). In this study, we highlighted eight identifiable hybrids from a total of 12 samples with high LH (>98.5%) and high AD (>2%) (Table 2; Appendices S2, S7). We focused on these eight because the other candidates may not have had progenitor lineages represented in our sample, and/or had lower sequence recovery. In the case of ×*Christelliopsis palmeri*, we investigated a relatively well-studied sterile F_1 hybrid nothogenus supported by chromosomal evidence, confirming the conclusion of Wagner (1993) that it represents an intergeneric hybrid. In all other examples we highlighted, the putative hybrids have long been recognized as stable species, representing both deep hybridization (an event occurring between two deeply diverged lineages) and ancient hybridization sensu Stull et al. (2023; a hybridization event resulting in a stable novel lineage). In the case of *Pelazoneuron kunthii*, a common and widespread taxon, we confirmed an earlier hypothesis, based on chromosomal evidence, that the taxon is of hybrid origin (Smith, 1971a). This example represents a pattern within the larger christelloid clade: two lineages in particular—the pantropical species *Christella hispidula* and the neotropical *Pelazoneuron* clade—have contributed a majority of putative hybrid taxa, evidently representing four independent hybrid speciation events. In three of these cases, *Pelazoneuron* was the maternal parent contributing its plastome, while the reciprocal parentage contributed to the origin of *Christella conspersa* (Tables 2, 3; Appendix S7; Patel et al., 2019).

In the case of *Christella chaseana* and *C. gueintziana*, these taxa may have resulted from a single hybridization event followed by speciation, or from two independent hybridizations involving the same or closely related progenitors—*Christella hispidula* and a member of the *Pseudocyclosorus* clade. *Sphaerostephanos elatus* was resolved to have ancestry in both the *Abacopteris* and *Pseudocyclosorus* clades (results consistent with those of Patel et al., 2019 and Fawcett et al., 2021), although its exact

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parentage is unclear, and sampling for both clades is limited, especially in Africa and Madagascar.

In the cases where alleles mapped to constituent genomes in a ratio of nearly 1:1 (Table 2), we suspect the hybrid is of relatively recent allotetraploid origin and that the reference clades are good representatives of progenitor lineages (e.g., *Pelazoneuron kunthii*). In contrast, we would potentially observe signals of subgenome dominance and lower proportions of mapped reads in a more ancient hybrid (Alger and Edger, 2020). In examples closer to a 2:1 ratio of read mapping (e.g., *Christella conspersa*), we believe one reference clade (corresponding to *Christella hispidula*) is very closely related to the progenitor lineage, but the second reference clade (*Pelazoneuron ovatum* var. *lindheimeri*) is more distantly related to the other progenitor of this allopolyploid taxon (likely an unsampled species of *Pelazoneuron*). In addition to biases from bait hybridization, deviations from a 1:1 allelic ratio could alternatively be explained by incomplete lineage sorting, hybrid backcrossing, gene duplication and losses, or ploidy differences between progenitors (Mallet et al., 2016; Nauheimer et al., 2021; Bloesch et al., 2022).

We focused on eight of the total 12 samples in our focal clade with high LH (>98.5%) and high AD (>2%) that also mapped to more than one reference clade (Tables 2, 3; Appendix S2). Based on their high LH and high AD, it is possible that the other four taxa are also of hybrid origin (*Amblovenatum opulentum*, *Christella hispidula* accession from Ethiopia, *Christella hispidula* accession from Mexico, and *Pneumatopteris afra*), but they are not well supported (Appendix S7). Before drawing conclusions about hybrid origins, we carefully evaluated each sample. For example, the *Christella hispidula* accession from Ethiopia has the highest LH and AD in all our studied samples. However, we opted to remove this sample from the data set due to its low sequence recovery (only 56.5% sequence recovery in the supercontig data set), as suggested by Nauheimer et al. (2021). This low sequence recovery may be due to the use of older herbarium tissue, which often have degraded DNA. We suspect that the observed high LH and AD values could sometimes simply be a consequence of sequence noise due to poor DNA library quality, as seen in this sample. Apart from high LH and AD values, the proportion of reads mapping to reference clades is also crucial for detecting hybrids. The aforementioned sample had 25.1% of reads mapping to the *Christella hispidula* clade reference, as expected, while the next closest match was 5.2% of reads mapped to the *Pelazoneuron* clade reference. In contrast, *P. kunthii* mapped to the same clade references (its presumed progenitors) at a rate of 26.8% and 27.2%, respectively (Appendix S2). All geographical and morphological evidence should also be considered before drawing conclusions, and high-quality data are needed to verify the hybrid status of any sample with confidence. In the case of *Pneumatopteris afra*, we did not find a clear signal of hybrid origin, although phylogenetic analyses (Fawcett et al., 2021) have resolved this morphologically distinctive taxon variously within *Abacopteris* and *Christella*. Earlier work has reported hybrids between this taxon and *Christella dentata*, *C. hispidula*, and a member of the pseudocyclosoroid clade (Viane, 1985). Further study and greater sampling of the christelloids in tropical Africa and Madagascar are needed to resolve the problem.

Cases of deep hybridization have been detected in several angiosperm families (e.g., Wu et al., 2010; Desjardins et al., 2015; Anghelescu et al., 2021; De Araujo et al., 2021; Díaz Lifante et al., 2023) as well as in bryophytes (Patel et al., 2023). It appears that the availability of more extensive genome data sets and the relatively smaller genome size in seed plants have facilitated the revelation of intergeneric, intertribal, and even higher taxonomic level hybridizations, such as between *Microbiota* and *Platycladus* of Cupressoideae (Liu et al., 2022), *Dasymaschalon* and *Friesodielsia* of Annonaceae (Guo et al., 2018), tribe Maleae and tribe Gillenieae of Rosaceae (Hodel et al., 2021), and Cornales and Gentianidae in asterids (Stull et al., 2020). We believe that a more extensive exploration of phylogenomics in ferns will likely uncover additional signals of ancient hybridization, which could contribute significantly to our understanding of fern evolution.

Challenges with phasing and age estimation

Hyb-Seq has been widely embraced as a tool for plant phylogenomics, facilitated by well-established probe sets targeting various land plant lineages (Johnson et al., 2019; Breinholt et al., 2021). This strategy is especially useful for tackling samples with highly degraded DNA that is usually found in historical but systematically important collections in herbaria. Earlier studies reported no significant difference in the number of loci recovered between herbarium tissue and silica-dried tissue, and no correlation between the age of the specimen and number of loci recovered (Fawcett et al., 2021). In our samples, tissue from herbarium specimens yielded an average of 380 loci and 241,724 nucleotides vs. an average of 385 loci and 308,005 nucleotides for silica-dried tissue; the silica-dried samples yielded significantly more nucleotides ($t = -6.8$; $P < 0.00001$).

One challenge is that the short-read approach of Hyb-Seq limits phylogenomic performance on hybrids and polyploids. When multiple moderately divergent copies are present at a single locus, a routine assembling strategy is problematic for inferring haplotigs (contig sequences representing different alleles) because of short-read lengths due to highly degraded DNA or limits of the sequencing platform (Saada et al., 2022). Phasing approaches are therefore applied at various stages to aid haplotig assembly. HybPhaser (Nauheimer et al., 2021) was developed to infer phylogenomics for allopolyploids and hybrids, particularly for plants. As used in this study, inclusion of parental lineages of allopolyploids or hybrids as references is indispensable for phasing reads into different genomic origins before haplotig assembly. More recently, other promising phasing strategies have been developed to deal with short-read data from Hyb-Seq that do not rely on

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reference sequences (e.g., Xie et al., 2016; He et al., 2018; Baaijens and Schönhuth, 2019; Moeinzadeh et al., 2020; Saada et al., 2021) and some of these have been applied in phylogenomics of plant polyploids and hybrids (e.g., Tiley et al., 2021; Mendez-Reneau et al., 2023). These phasing approaches are based on the read mapping profile during assembly, which then generates phased haplotigs for each locus. Because the resulting haplotigs remain unsorted as to their genomic origins, the phylogenomic inference relying on a concatenated matrix needs assistance from an additional phasing process, such as homologizer (Freyman et al., 2023). We look forward to future studies that incorporate comprehensive tests to compare the performance of these different phasing strategies (e.g., Saada et al., 2022). A Hyb-Seq approach using long-read sequencing is an alternative promising solution; increasing informative SNPs per read across distant sites would enhance our ability to correctly assemble haplotigs. Such an approach has been showcased in targeted assembly of gene families of some crops (e.g., Giolai et al., 2017). An application for phylogenomic purposes is worth developing.

Potential concerns with Hyb-Seq data include sequence purity and missing characters in the output matrix. The inclusion of paralogs, high numbers of gene copies (e.g., resulting from hybrids and polyploids), erroneous bases potentially resulting from increased PCR cycles, and low read coverages may all impact data quality (Aird et al., 2011; Fountain et al., 2016; Andermann et al., 2020). When a considerable proportion of reads from paralogs or different copies are retained and merged into the final assembly, misassembled contigs with erroneous bases can subsequently lead to increased branch lengths and misleading topologies in resulting trees. In addition, when read coverage is insufficient, the concatenated matrix will have a larger proportion of missing data. In the present study, we found several such cases, particularly for herbarium samples, including *Abacopteris* aff. *peltochlamys*, *Pseudocyclosorus repens*, *Trigonospora obtusiloba*, *Christella hispidula* (accessions from Ethiopia, Java, and Mexico), *Christella hispidula* var. *inconstans*, *Pelazoneuron patens* var. *dissimile*, and *Pakau pennigera* (Figure 1). These samples are represented by extraordinarily long terminal branch lengths compared with other samples, and these accessions also had the most missing data with 117,487–175,695 bp in our initial phylogenomic data set compared with the average of 318,903 bp. When performing treePL dating on such a phylogenomic data set, the inferred ages from resulting chronograms are likely overestimated (data not shown), which are nearly double the previous estimates (e.g., Nitta et al., 2022). The removal of these taxa, introns, and some exons with a high rate of missing data highly reduced the overall proportion of missing data from 17% to 1.7%. With the more filtered matrix, we inferred a chronogram with divergence times that fall into an age range consistent with our current understanding of the fern tree of life (Testo and Sundue, 2016; Nitta et al., 2022; Table 3). In addition, we found inferred ages could significantly differ depending on the partitions/substitution models used (Appendix S4). Apparently, the parameter-rich model (e.g., every partition with a different GTR + F + R10 model) could result in a greater age estimate compared to that from the simpler model (e.g., a partition scheme by ModelFinder or "codon-property"; Table 3; Appendix S4). These findings highlight the need to cautiously interpret age estimates and suggest the further need to improve the missing data filtering, supercontig assembly methods, and substitution model selections for Hyb-Seq data to obtain a more reliable inference of branch length for estimating divergence age.

Analyses of ecological niche differences

The ecological niche of the allopolyploid *Pelazoneuron kunthii* had a larger degree of overlap with each of its progenitor species (*C. hispidula*, $D = 0.400$; *P. ovatum*, $D = 0.445$) than its progenitors did with each other ($D = 0.286$) (Table 4). This result is consistent with the pattern most frequently observed in other allopolyploid–diploid progenitor niche analyses (Marchant et al., 2016). Our results showed divergence (significant niche equivalency test), but no significant niche similarity between the niches of *Christella hispidula* and each of the other two species. These differences may be in part explained by geography and related ecological conditions. For example, *C. hispidula* is widespread across paleotropical and neotropical biomes, while *P. ovatum* is restricted to Central America and the southeastern United States. Differences in geographic range may also explain the unusually high (21.3%) pairwise unfilling (occupied niche in *C. hispidula* not found in *P. ovatum*). However, despite the observed niche divergence, these taxa have been observed to grow in close proximity, and hybridization between the two phylogenetically divergent progenitors may have been facilitated by paleoclimatic stress or disturbance events within their shared geographical ranges. The evolution of allopolyploid hybrids is frequently associated with secondary contacts between progenitors following major disturbance in deglaciated areas (Brochman et al., 2004; David, 2022). In this case, the localities where all three taxa co-occur in Florida are less than 10 m above sea level and were repeatedly inundated by seawater and exposed during Pleistocene glacial cycles (Hine et al., 2017). This disturbance may have provided opportunity for repeated hybridization of two progenitors and the establishment of allopolyploid progeny (Fowler and Levin, 2016). Further, selection processes from disturbance or stress conditions may have also created new niches into which the hybrid progeny colonized and adapted (Schluter, 2009; Barrington, 2020). The moderate overlap ($D = 0.4$), high stability, and low expansion and unfilling observed between the progenitor, *C. hispidula*, and its descendant, *P. kunthii*, is consistent with the hypothesis of shared ecological similarity due to inheritance. The sympatric geographical distribution and overlapping habitats of *Pelazoneuron kunthii* and *P. ovatum* (Central America and southeastern United States) may have contributed to their significant niche similarity, indicating conservatism. The higher

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niche overlap between them can be decomposed into 77% niche stability, 23% expansion, and 10% unfilling, which suggests niche partitioning may be the result of microecological variation in occupied ranges (Baniaga et al., 2020).

Pelazoneuron kunthii, the allopolyploid taxon studied here, is a highly successful native species, based on the relative abundance of occurrences documented in iNaturalist records and herbarium specimens compared to its progenitors (GBIF.org, 2023). Although *P. kunthii* occurs sympatrically with its progenitors (Figure 2A–C), it is more frequent than *C. hispidula* in habitats impacted by anthropogenic disturbance, such as roadsides and openings (Smith, 1971b), and *P. kunthii* is evidently expanding its range. For example, the number of occurrences recorded in Arkansas jumped from a single report in 1975 to records representing 26 counties by 2011 (Peck, 2011). It also was recently reported as naturalized in California (*Riefner 14-83*, UC), Spain (López-Tirado, 2016; Ranchal Muñoz et al., 2021), Israel (*Leschner s.n.*, UC), and Palestine (*Mendelssohn s.n., Yaffe s.n.*, UC), presumably escaping from cultivation. This rapid range expansion underscores the potential of polyploid taxa to adapt to novel environments and potentially develop weedy or invasive ecological behavior (te Beest et al., 2012; Moura et al., 2021). The ecological success of allopolyploid taxa in stressful environments is reported in ferns (Barrington, 2020) and in flowering plants, amphibians, insects, and ray-finned fishes (Feldman and Levy, 2009; Shimizu-Inatsugi et al., 2017; David, 2022). Success as a weed has been well documented in the pantropical allopolyploid *Christella dentata*, which, since its recent introduction in the Americas, is rapidly spreading and may be outcompeting a native diploid member of its progenitor lineage, *C. hispidula*, in areas where their ranges overlap (Strother and Smith, 1970). Allopolyploids may have higher evolutionary potential than their progenitors due to increased genetic diversity contributed by both parents, and the associated capacity for phenotypic plasticity or divergence, which may confer higher stress tolerance and the ability to occupy various environments (Godfree et al., 2017; Van de Peer et al., 2017; Wei et al., 2019).

Nomenclatural considerations for intergeneric hybrids

Traditions for naming intrageneric hybrids (Wagner, 1969) and nothogenera (sterile intergeneric hybrids) are well established, the latter involving a portmanteau of parental genus names (Turland et al., 2018). In contrast, there is no consensus for conventions on naming stabilized lineages of intergeneric hybrid origin. Adhering to a concept of strict monophyly would potentially necessitate a novel genus for each stable taxon resulting from intergeneric hybrid evolution, which is further complicated by the likelihood of multiple independent origins of hybrid taxa. There is precedent for the recognition of a novel genus representing multiple species arising from independent intergeneric hybridization events involving the same two genera. For example, *Stebbinsoseris* is a genus of two allotetraploids involving parentage by *Uropappus lindleyi* and different species of *Microseris* (Jansen et al., 1991). In the present study, we adopt the previously published combinations used by Fawcett and Smith (2021), tentatively treating intergeneric hybrid species within the genus of the maternal progenitor with which they share a plastome (Stull et al., 2023). Although neither of these is an ideal solution, we presently follow the least taxonomically disruptive course.

CONCLUSIONS

Hybridization plays an essential role in plant evolution and is especially important in ferns. Ancient genome duplication events, frequent allopolyploid speciation among congeners, and the capacity for deep hybridization have all been well documented. We presented evidence for at least six instances of intergeneric hybrid speciation between lineages estimated to have diverged 20 million or more years ago. In one such case, we highlighted an example of a widespread and abundant intergeneric allopolyploid with an ecological niche that overlaps with both of its progenitors, while showing a capacity for expansion into novel habitat, with demonstrated weedy tendencies. We suggest that the christelloid clade represents an excellent system for the study of hybridization due to its amenability to cultivation and experimental breeding, and for its unusual capacity for hybridization. One species in particular, *Christella hispidula*, is especially capable of hybridizing. As a constituent genome of the tetraploid *Christella dentata*, it has contributed to a disproportionate number of sterile intergeneric hybrids and also acted as a diploid progenitor in five of the six intergeneric hybrid species highlighted here. Our findings suggest that focusing on other lineages prone to frequent infra- and intergeneric hybridization may reveal other examples of deep hybrid speciation.

Our ability to recognize hybrid taxa resulting from crosses between deeply divergent, species-rich, and morphologically diverse lineages has been limited, especially when compared to the relative ease of recognizing recent F_1 hybrids by identifying intermediates occurring near progenitors, e.g., *Polystichum californicum* (Wagner, 1973). Our results highlight the possibility that hybrid speciation involving deeply divergent lineages may be much more common than previously thought. The preponderance of traditional methods relying on strictly bifurcating phylogenetic trees to represent evolutionary relationships has also impacted our ability to frame evolutionary hypotheses in ways that allow for reticulation. New methods capturing larger portions of genomes and phasing alleles allow us to recognize previously overlooked diversity, reticulate ancestry, and evolutionary complexity. We should strive to take a more nuanced look at the tangled branches of the tree of life.

AUTHOR CONTRIBUTIONS

All authors contributed equally to the manuscript.

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ACKNOWLEDGMENTS

We are indebted to Alan Smith (UC) whose pioneering work on the Thelypteridaceae enabled this research. We are grateful to the GoFlag (Genealogy of Flagellate Plants) Consortium supported by NSF DEB-1541506 for providing a wealth of publicly available sequence data and especially to Gordon Burleigh and Emily Sessa (FLAS, NY) for their roles in generating these data. We thank Bruce Baldwin (JEPS), Weston Testo (VT), and two anonymous reviewers for helpful comments on the manuscript. We thank Michael Barker, Kelsey Glennon, Agnieszka Golicz, Yuannian Jiao, Michael McKain, and Ya Yang for organizing the special issue on polyploid biology.

DATA AVAILABILITY STATEMENT

Data are available on Dryad: https://datadryad.org/stash/share/IZEve5_3OoOPT-RveVLxVMsu_DnIvE_z2CgvIOb-5_U (Tseng et al., 2024); previously published sequence data are available on Dryad doi.org/10.5061/dryad.gxd2547j4 (Fawcett et al., 2022) and NCBI Sequence Read Archive Bioproject 646399 (https://www.ncbi.nlm.nih.gov/bioproject/?term=646399).

ORCID

Yu-Hsin Tseng http://orcid.org/0000-0002-8166-5690
Li-Yaung Kuo http://orcid.org/0000-0002-3388-3757
Israel Borokini http://orcid.org/0000-0002-1258-7932
Susan Fawcett http://orcid.org/0000-0001-5021-1779

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Appendix S1. List of voucher specimens used in phylogenetic analyses (.csv file)

Appendix S2. Scatterplot representing the locus heterozygosity (LC) and allele divergence (AD) across the samples. Each data point corresponds to an individual sample, with two categories indicated by labels "Phased sample (P)" and "Clade reference (R)"

Appendix S3. Phylogenetic tree of christelloid clade reconstructed using maximum likelihood analysis based on GoFlag 408 loci sequence data. Major clades within the tree are designated as Clade A to Clade H. Clade references are denoted by blue colors. Numbers above branches indicate bootstrap values of maximum likelihood (MLBS)/gene concordance factors (gCF)/site concordance factor (sCF)

Appendix S4. Divergence estimates and partition models used in dating analysis (.xlsx file)

Appendix S5. Georeferenced specimens used for niche analysis (.csv file)

Appendix S6. Heat map showing recovery efficiency for 451 GoFlag loci in the christelloid clade, assembled by HybPiper. Columns represent loci and rows are samples. Colors indicate sequence recovery percentage relative to reference gene length

Appendix S7. Sequence length, AD, LH, and result of percentages of reads matching to each reference in our sampling. Clade reference and phased sample are noted. Red highlights indicate higher percentages of unambiguous read mappings to the reference in phased samples

How to cite this article: Tseng, Y.-H., L.-Y. Kuo, I. Borokini, and S. Fawcett. 2024. The role of deep hybridization in fern speciation: Examples from the Thelypteridaceae. *American Journal of Botany* 111(8): e16388. https://doi.org/10.1002/ajb2.16388

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